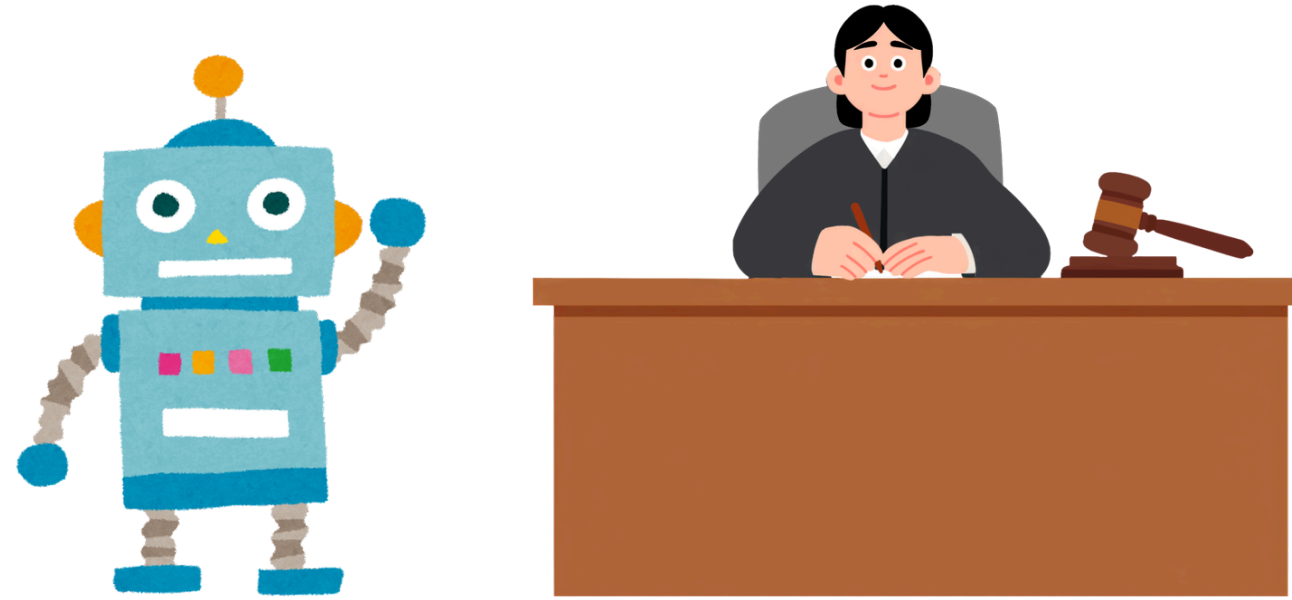




LLM-as-a-Judge ^_^

Introductory Lecture

Thura Aung





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@ June

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Education

B.Eng, Software Engineering (Artificial Intelligence Track),
King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand.
Child's Dream Foundation's University Scholarship (Fully-funded).

2023 - Present

Research Experience



AI SINGAPORE[®]

looloo

Language
Understanding
 $\lambda(\mu)$ Laboratory



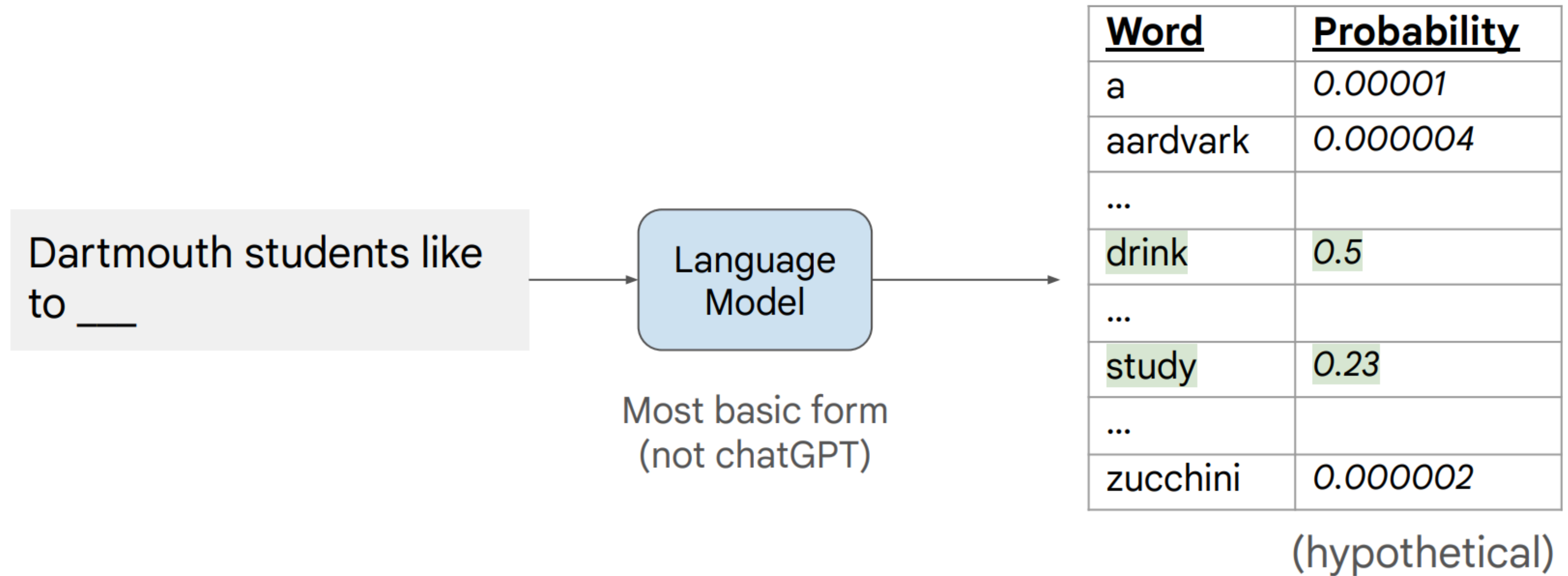
NUS
National University
of Singapore

The lecture is highly influenced by the following lectures

- <https://ai4comm.media.mit.edu/>
- <https://cme295.stanford.edu/>

Large Language Models

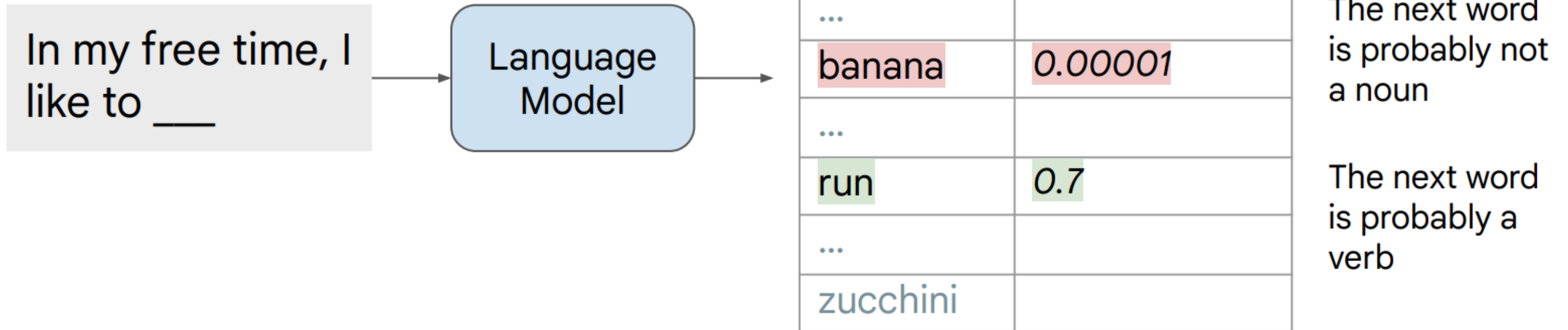
LLM



LLM

[1/8] What do language models learn from next word prediction?

→ Grammar

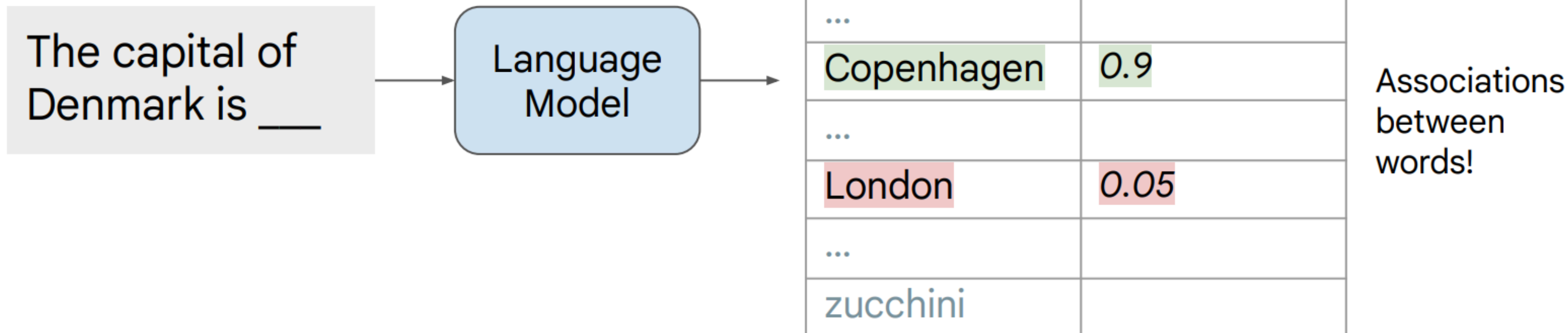


(hypothetical)

LLM

[2/8] What do language models learn from next word prediction?

→ Facts about the world



(hypothetical)

LLM

[3/8] What do language models learn from next word prediction?

→ Lexical semantics

I went to the
zoo to see
giraffes, lions,
and

Language
Model

<u>Word</u>	<u>Probability</u>
a	
...	
spoon	0.00001
...	
zebras	0.6
...	
zucchini	

The next word
is probably
related to
giraffes and
lions

(hypothetical)

LLM

[4/8] What do language models learn from next word prediction?
→ Sentiment analysis

I was engaged
and on the edge
of my seat the
whole time. The
movie was ____

Language
Model

<u>Word</u>	<u>Probability</u>
a	
...	
bad	0.1
...	
good	0.9
...	
zucchini	

Well, "engaged" is
pretty indicative
of a positive
sentiment

(hypothetical)

LLM

[5/8] What do language models learn from next word prediction?
→ Harder sentiment analysis

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was —

Language Model

<u>Word</u>	<u>Probability</u>
a	
...	
bad	0.7
...	
good	0.3
...	
zucchini	

Some more-complex understanding needed

(hypothetical)

LLM

[6/8] What do language models learn from next word prediction?

→ Translation

The word for
"pretty" in
Spanish is ____

Language
Model

<u>Word</u>	<u>Probability</u>
a	
...	
bonita	0.8
...	
hola	0.03
...	
zucchini	

Understanding
of multiple
languages

(hypothetical)

LLM

[7/8] What do language models learn from next word prediction?
→ Spatial reasoning

Iroh went into the kitchen to make some tea. Standing next to Iroh, Zuko pondered his destiny. Zuko left the ____

Language Model

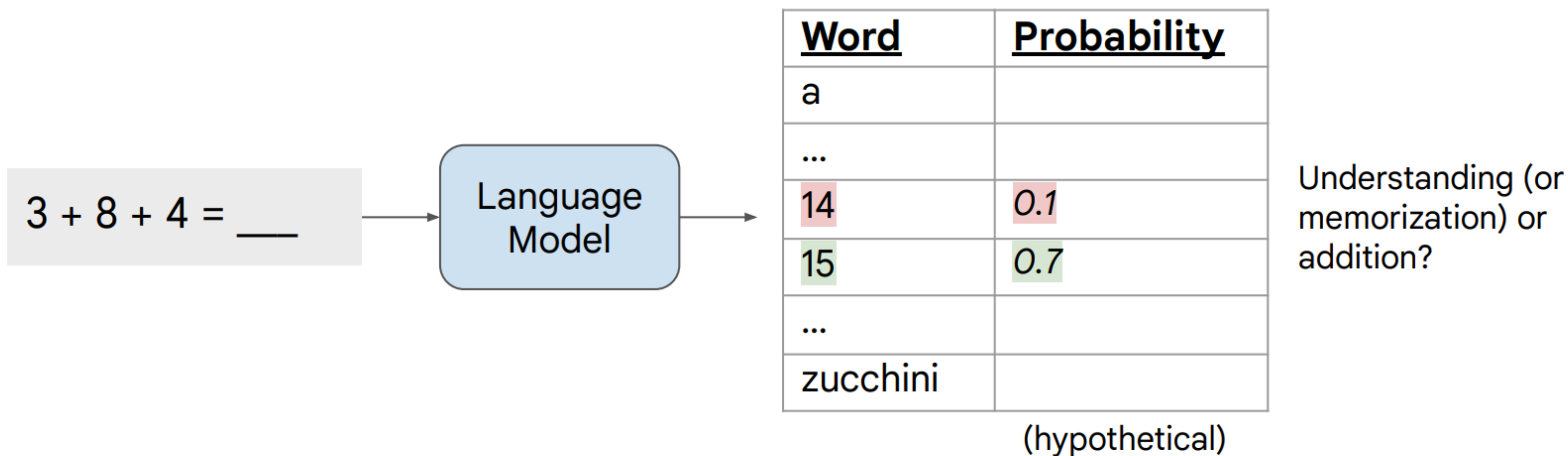
<u>Word</u>	<u>Probability</u>
a	
...	
...	
kitchen	0.8
...	
zucchini	

(hypothetical)

LLM

[8/8] What do language models learn from next word prediction?

→ Easy arithmetic



LLM

[1/6] What can't language models do from next word prediction?
→ Not current world knowledge

The stock price
of APPL on
March 1st, 2023
is ____

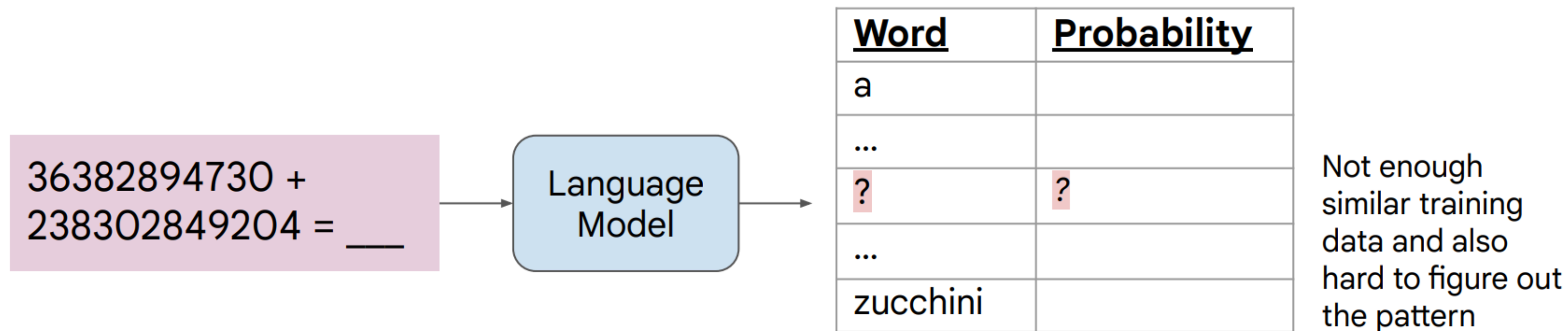
Language
Model

<u>Word</u>	<u>Probability</u>
a	
...	
?	?
...	
zucchini	

Language model
was trained
before that date

LLM

[2/6] What can't language models learn from next word prediction?
→ Not arbitrarily long arithmetic



[3/6] What can't language models do from next word prediction?
 → Math questions you probably wouldn't be able to do

Take the nineteenth digit of Pi and multiply it by the e to the fourth power. The resulting ones-digit of the resulting number is ____

Language Model

<u>Word</u>	<u>Probability</u>
a	
...	
?	?
...	
zucchini	

A lot of intermediate steps so hard to figure out the pattern

LLM

[4/6] What can't language models do from next word prediction?

→ Predict the future

The winner of the
FIFA world cup in
2026 is ____

Language
Model

<u>Word</u>	<u>Probability</u>
a	
...	
?	?
...	
zucchini	

Nobody can do
this

LLM

[5/6] What can't language models do from next word prediction?
→ Information that isn't in the training data

Jason Wei's favorite
color is ____

Language
Model

<u>Word</u>	<u>Probability</u>
a	
...	
?	?
...	
zucchini	

That information
is probably not
seen in its training
data

LLM

[6/6] What can't language models do from next word prediction?
→ Can't take in extremely long pieces of text

[2,000 page Harry Potter fan-fiction]
What happened after Harry opened the chest for the second time? ____

Language Model

<u>Word</u>	<u>Probability</u>
a	
...	
?	?
...	
zucchini	

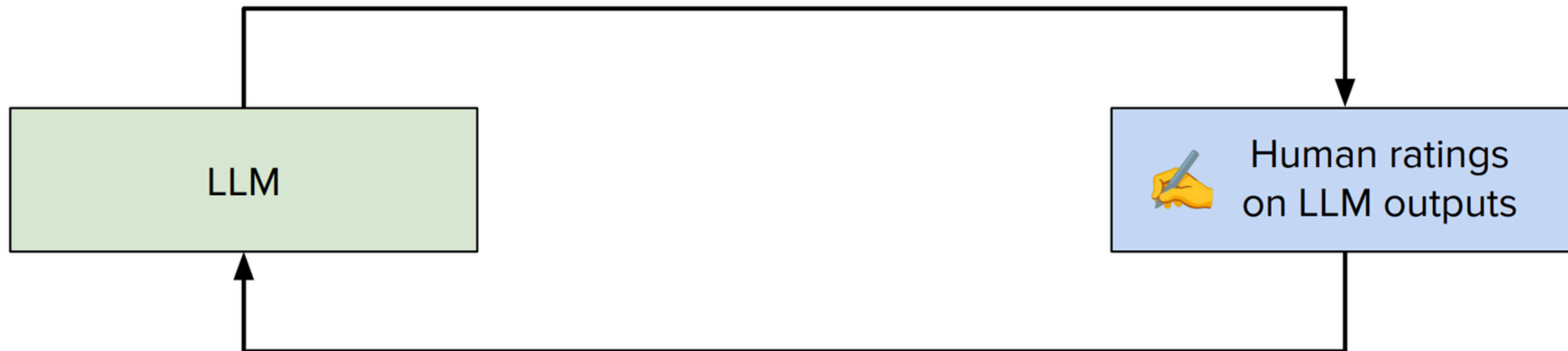
Can't take in very long inputs
(limited to like 4k words-ish)

Human Evaluation

Human Eval

Context. LLM generates free-form text which is hard to evaluate

Human rating. Closest to truth and "gold"



Human Eval

- A human (e.g. crowd turker) compares the model answer to the real answer.
- Typically asked to assess:
 - Coherence, readability, fluency
 - Grammaticality
 - Extent to which the model follows instructions
- Preference judgements:
 - Example: Choose the passage that is more [insert quality]
 - Could have a third option specifying that both passages are equally good.
- Rating a passage (e.g., Likert scale):
 - Example: Thinking about [insert assessed quality], rate the following passage on a scale of 1 to 5 with 1 being the worst and 5 being the best.
 - Example: The generated story follows the instructions (e.g., includes all characters). How much do you agree with this statement?

Human Eval



Prompt

What birthday gift
should I get?



Model response

A teddy bear is almost always a
sweet gift—just pick one that
feels right to you.



Criteria

Usefulness

Rater 1



Rater 2



Human Eval

Subjectivity of evaluation task

agreement rate = p_o ← "Observed"

Limitation. What if this was pure 🎲?

"Observed" → $\kappa = \frac{p_o - p_e}{1 - p_e}$ ← "Expected" if by chance

Metric. *"How much better is our agreement than what we'd expect just by chance, given how the raters actually use the categories?"*

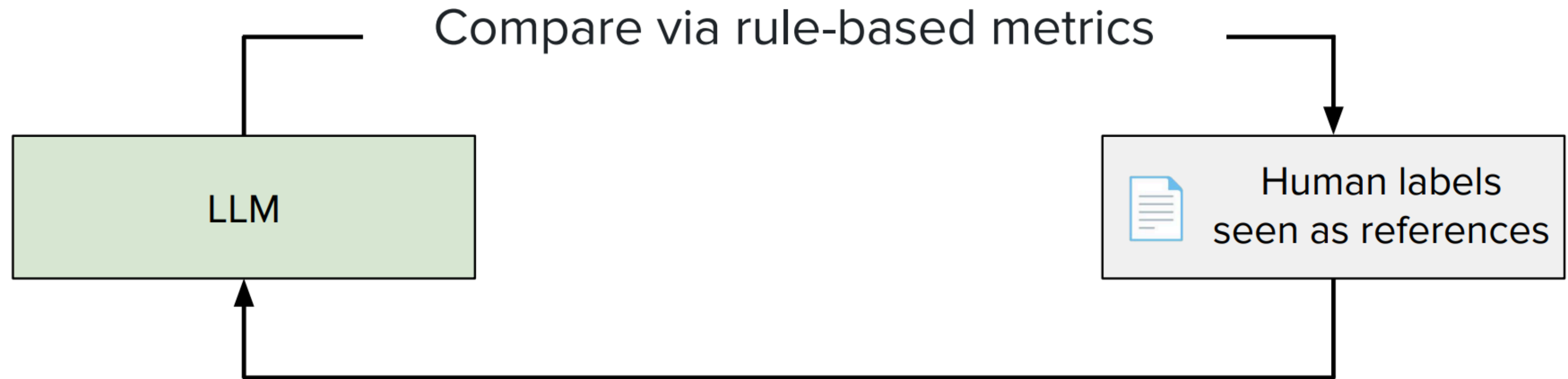
Variants. Cohen's Kappa, Fleiss' Kappa, Krippendorff's alpha

- Subjectivity of evaluation task
- Slow
- Expensive

Objective Evaluation Metrics

Objective Metrics

Idea. Write labels once, and use it as reference



Objective Metrics

METEOR. Metric for Evaluation of Translation with Explicit ORdering

$$\text{METEOR} = \boxed{F_{\text{mean}}} \times (1 - \boxed{p})$$

$$F_{\text{mean}} = \frac{P \times R}{\alpha P + (1 - \alpha)R}$$

$$p = \gamma \times \left(\frac{c}{u_m} \right)^\beta$$

- "METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments", Banerjee et al., 2005.
- "BLEU: A Method for Automatic Evaluation of Machine Translation", Papineni et al., 2002.
- "ROUGE: A Package for Automatic Evaluation of Summaries", Lin et al., 2004.

BLEU. BiLingual Evaluation Understudy

$$\text{BLEU} = \boxed{\text{BP}} \times \exp \left(\frac{1}{N} \sum_{n=1}^N \log(\boxed{p_n}) \right)$$

$$\text{BP} = \min \left(1, \exp \left(1 - \frac{n_{\text{ref}}}{n_{\text{pred}}} \right) \right)$$

$$p_n = \frac{\sum_{n\text{-gram} \in \hat{y}} \max_{k \in [1, m]} \left[\text{count}(\text{matching } n\text{-gram between } \hat{y} \text{ and } y_k) \right]}{\sum_{n\text{-gram} \in \hat{y}} \text{count}(n\text{-gram})}$$

Objective Metrics

- Does not take into account stylistic variations
- Correlation with human rating is not that great
- Still requires human ratings!

A plush teddy bear can comfort a child during bedtime.

Soft stuffed bears often help kids feel safe as they fall asleep.

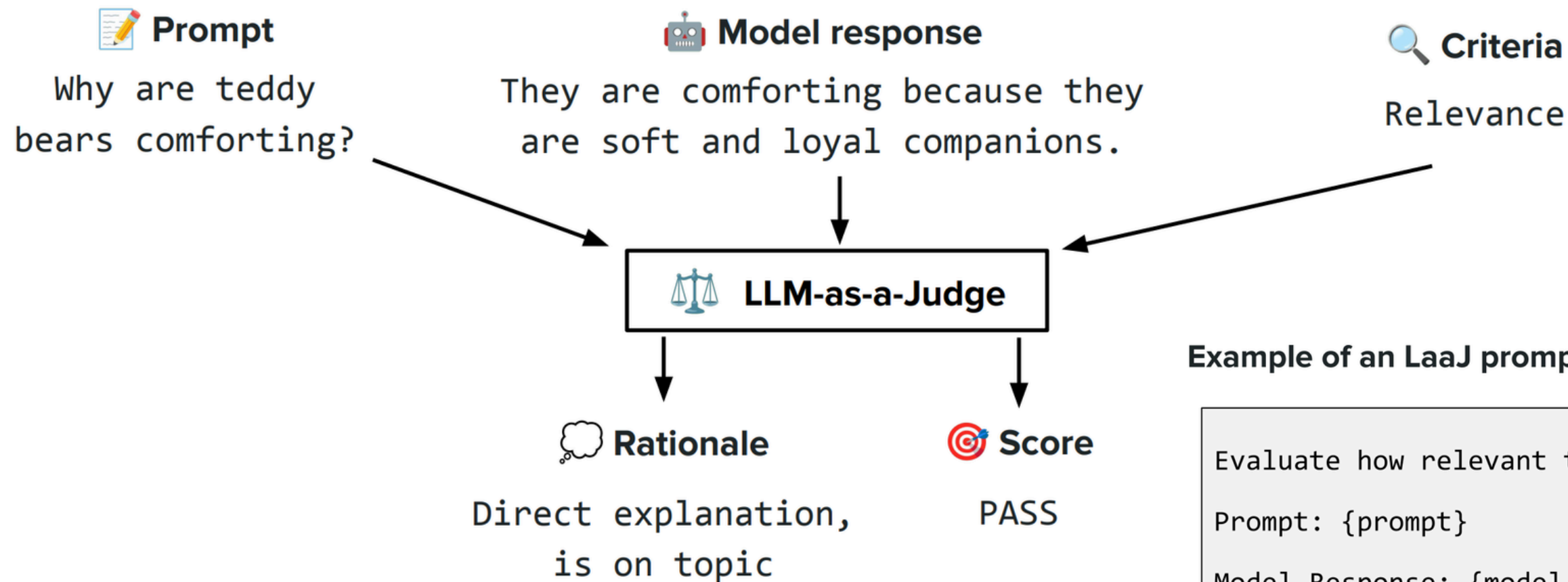
Many youngsters rest more easily at night when they cuddle a gentle toy companion.

**LLM-as-a-Judge
is subjective ?**

LaaJ

LaaJ = LLM-as-a-Judge

Idea. Use LLM to rate quality of response.



Example of an LaaJ prompt.

Evaluate how relevant the model's answer is to the user's prompt.

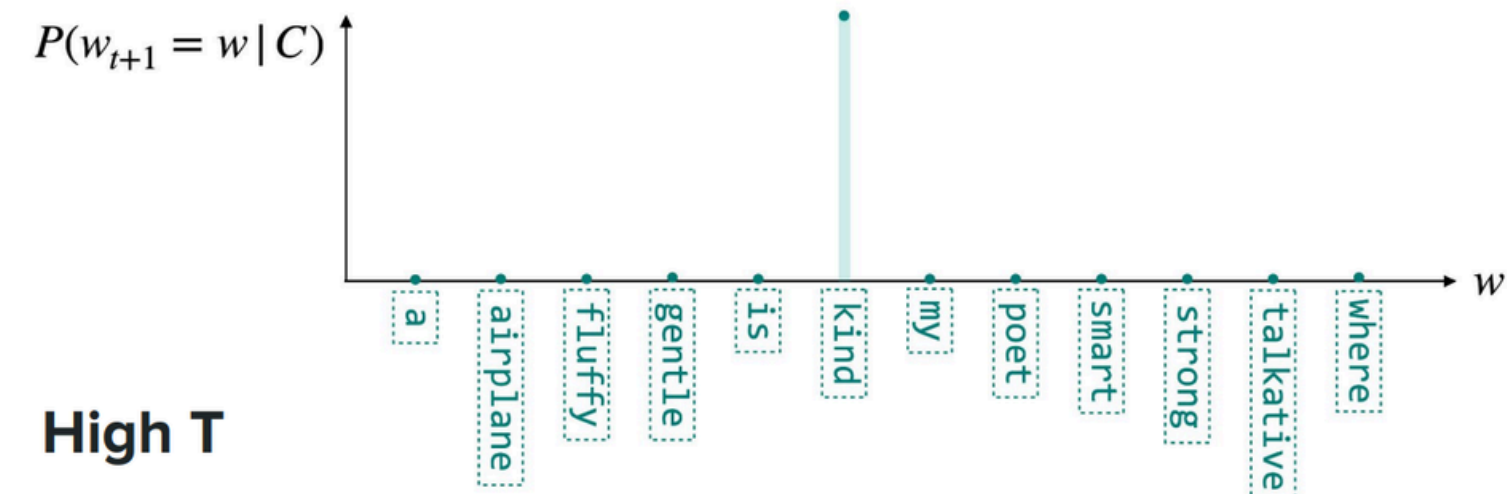
Prompt: {prompt}

Model Response: {model_response}

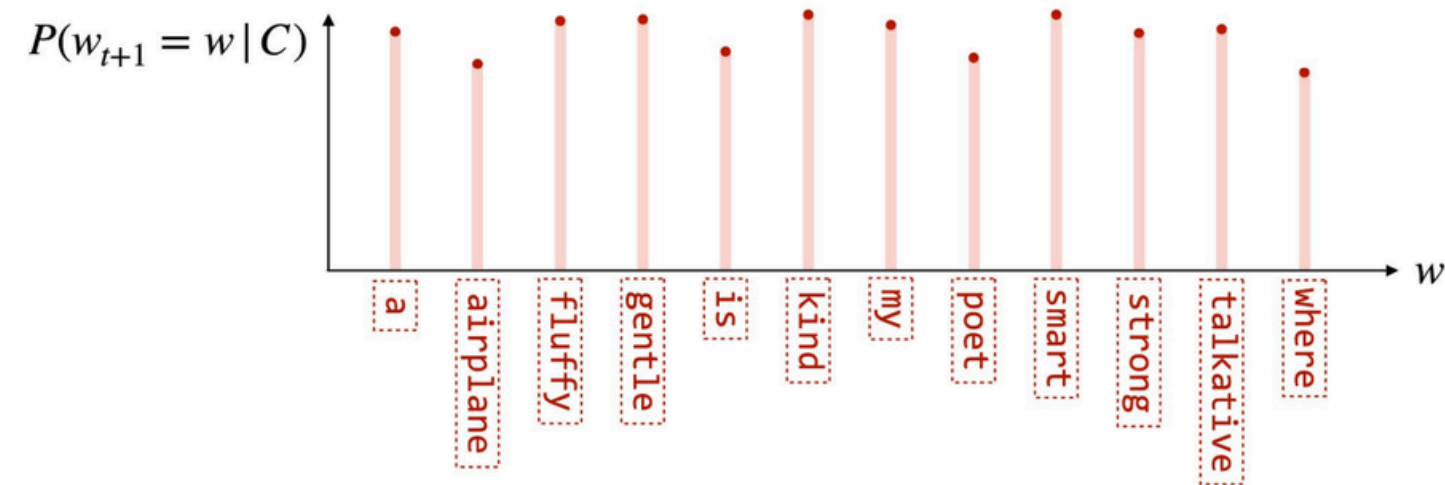
Return:

- Rationale (1-2 sentences)
- Score: 1 if mostly relevant, 0 if mostly irrelevant.

Small T



High T



Enforce output format in practice

Motivation. Generate output in a specific format

Input prompt

*Generate a description of my
33-year old teddy bear who
likes reading.*

Do this in JSON format.

Desired output (JSON)

```
{
  "first_name": "teddy",
  "last_name": "bear",
  "age": 33,
  "hobby": "reading"
}
```

LaaJ: Position Bias

Pointwise

Evaluate the quality
of:

Response



Very good

Pairwise

Which one is better:

Response A

or

Response B

?



Response A

Which one is better:

Response B

or

Response A

?



Response B

Remedy. "Take the average", or tweak position embeddings

"Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena", Zheng et al., 2023.

LaaJ: Verbosity Bias

Which one is better:

Response A

or

Response B

?



Response B

Response A

This response is short and correct.

Response B

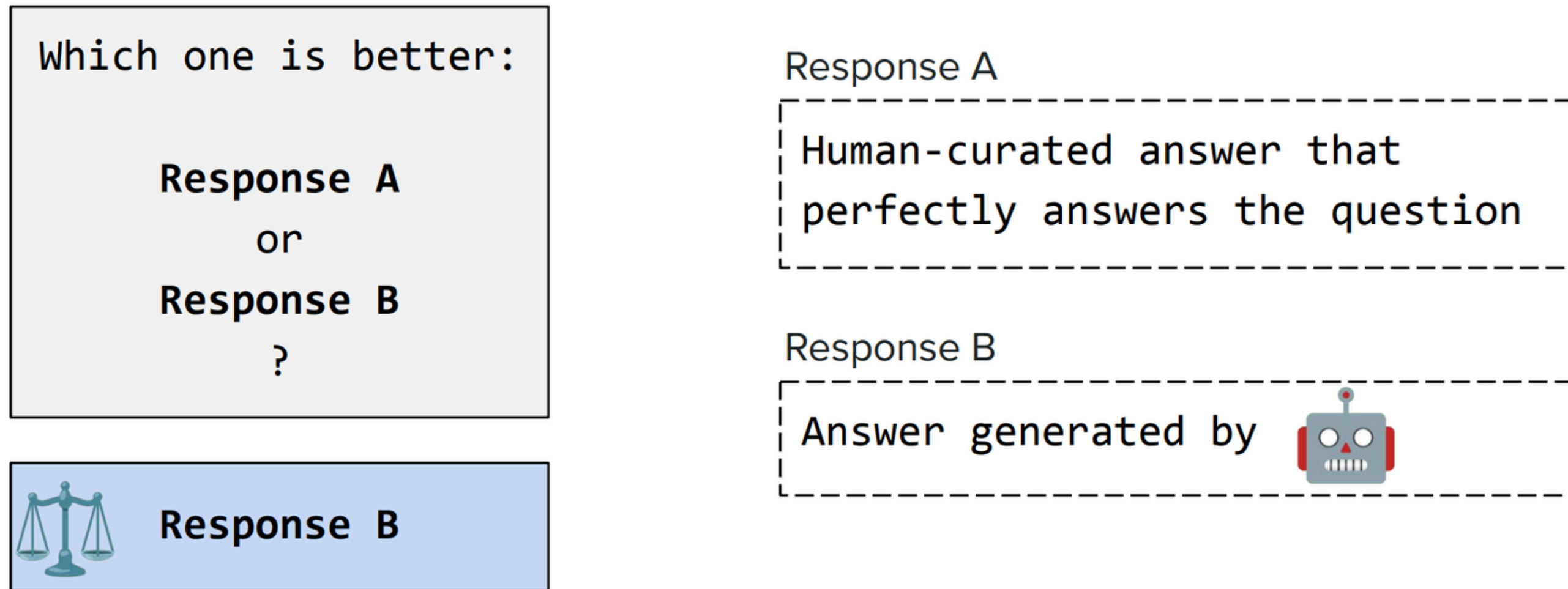
This response goes into details, a lot of which are not needed and is overall not very helpful.

Remedy. Explicit guidelines, few-shot, and/or penalty on output length

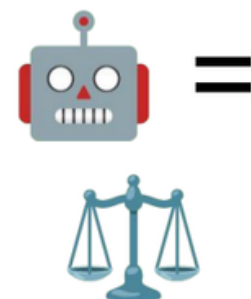
"Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena", Zheng et al., 2023.

LaaJ: Self-enhancement bias

AI evaluator to systematically assign higher scores to responses generated by its own model family



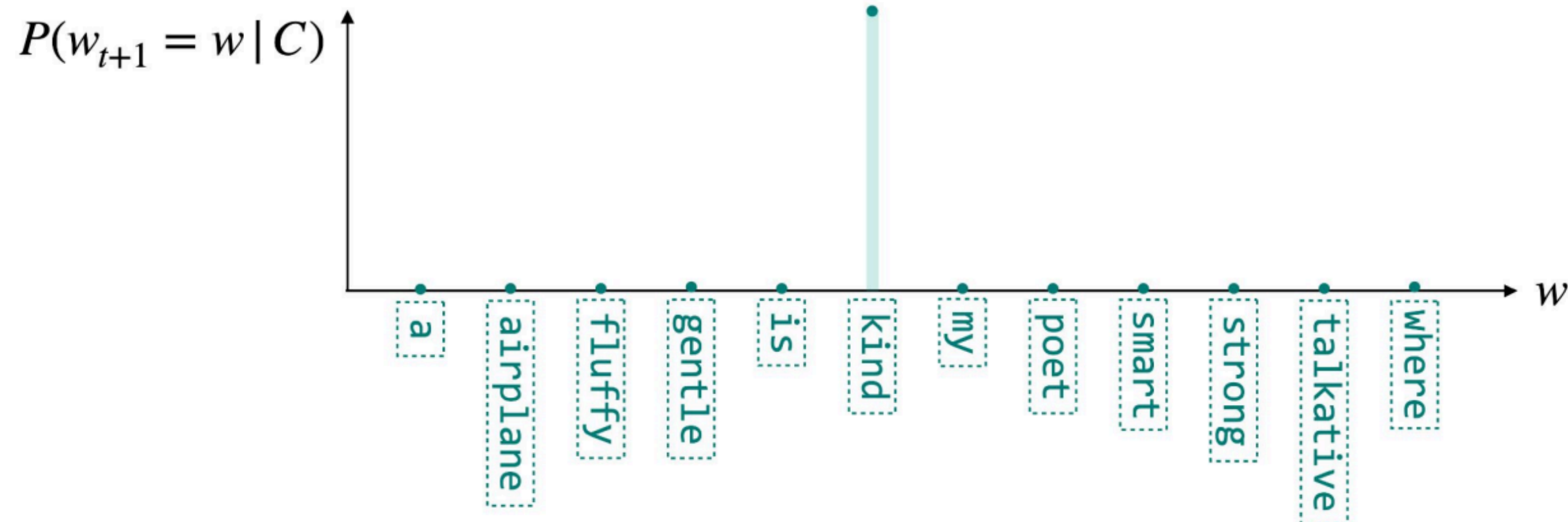
Remedy. Don't use



"Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena", Zheng et al., 2023.

LaaJ: Practices

- Mitigate biases (position, verbosity, self-enhancement, etc.)
- Calibrate with human judgements
- Low temperature for reproducibility
- Crisp guidelines
- Binary scale over more granular ones
- Write rationale before outputting score



LaaJ: LaaJ Metrics

Typical dimensions we want to evaluat

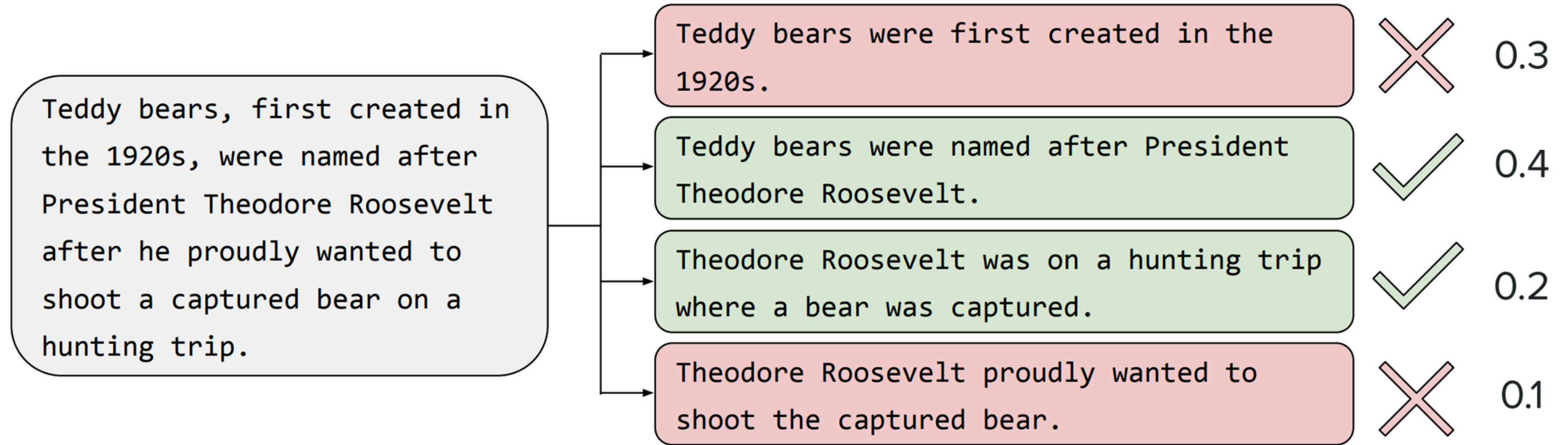
 **TASK PERFORMANCE**

- Usefulness
- Factuality
- Relevance

 **ALIGNMENT**

- Tone
- Style
- Safety

LaaJ: Fact Check



$$\text{score} = \sum_{i=1}^n \alpha_i \times \text{score}_i$$

with α_i importance of fact i

Directions

Eval Fairness & Bias

Definitions of Bias / Fairness

- Where models demonstrate unfair, discriminatory, or hateful behaviour
- This can be particularly harmful if targeted towards sensitive personal attributes, such as gender, sexuality, race and religion.
- Harms can arise even from “correct” or intended uses, depending on where and how they are deployed, and in predictive applications as well as generative ones.

Toxicity

Profanity

Sexually Explicit

Gender Bias

Sexual Orientation Bias

Ethnic/Cultural Bias

Hate Speech

Implicit Bias

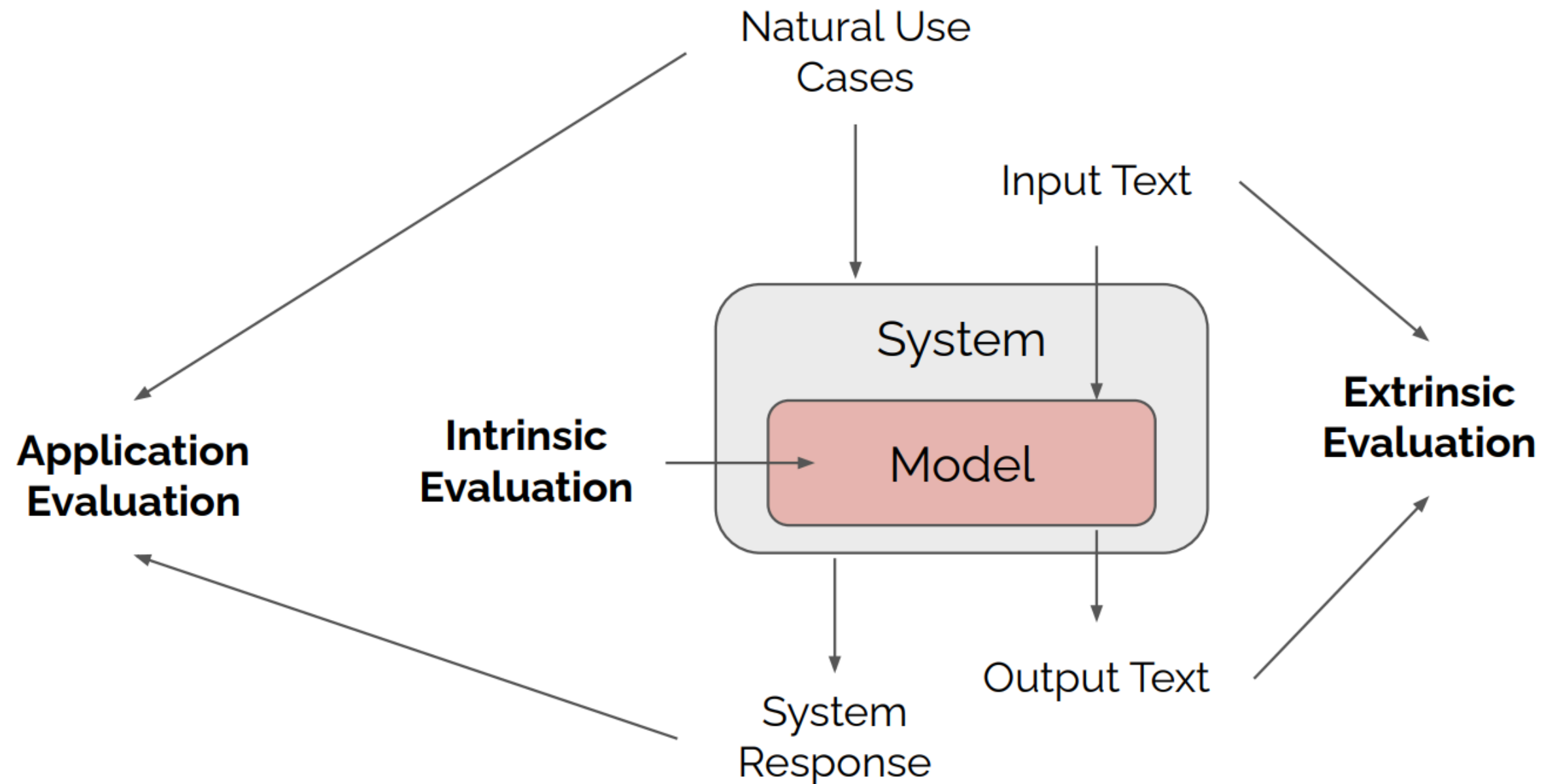
Discriminatory or Unfair

Social Impact

Eval Fairness & Bias

A Generative Language Model:

- Emulates text scraped from across the web
- Is often optimized for subsets of users (western, affluent, etc)



Eval Fairness & Bias

What is Bias in NLP Models?

- Where models demonstrate unfair, stereotyped, or discriminatory behaviour towards sensitive demographics, such as gender, sexuality, race and religion. How has prior work evaluated bias?
- **Intrinsic Bias** → Evaluating the inner state of the model itself ○ E.g. African-American names are more closely associated with unpleasant words in the model embedding space.
- **Extrinsic Bias** → Evaluating the behaviour of the model from (input, output) pairs ○ E.g. Given leading prompts, how often will the model generate a toxic, biased response?
- **Application Bias** → Evaluating the full system in the setting where it is deployed ○ E.g. Translation technologies systematically generating incorrect and stereotyped genders

Eval Fairness & Bias



Biased Generation

Benchmarks:

- Stereoset
- UNQover
- Bias Benchmark for QA (BBQ)
- Winogender

General Resources for NLP Bias + Fairness:

- [http://web.cs.ucla.edu/~kwchang/talks/emnlp19-fairnlp /](http://web.cs.ucla.edu/~kwchang/talks/emnlp19-fairnlp/) Chang et al. (2019)
- Language (Technology) is Power: A Critical Survey of "Bias" in NLP. Blodgett et al. (2020)
- <https://github.com/uclanlp/awesome-fairness-papers> Chance et al. (2023) o Recent list of related publications in ML Fairness
- Stereotyping Norwegian Salmon: An Inventory of Pitfalls in Fairness Benchmark Datasets. Blodgett et al. (2021) o Problems in common benchmarks

Winobias Toxicity

Identification

Benchmarks:

- Civil Comments
- Toxigen
- DynaHate

Agent-as-a-Judge

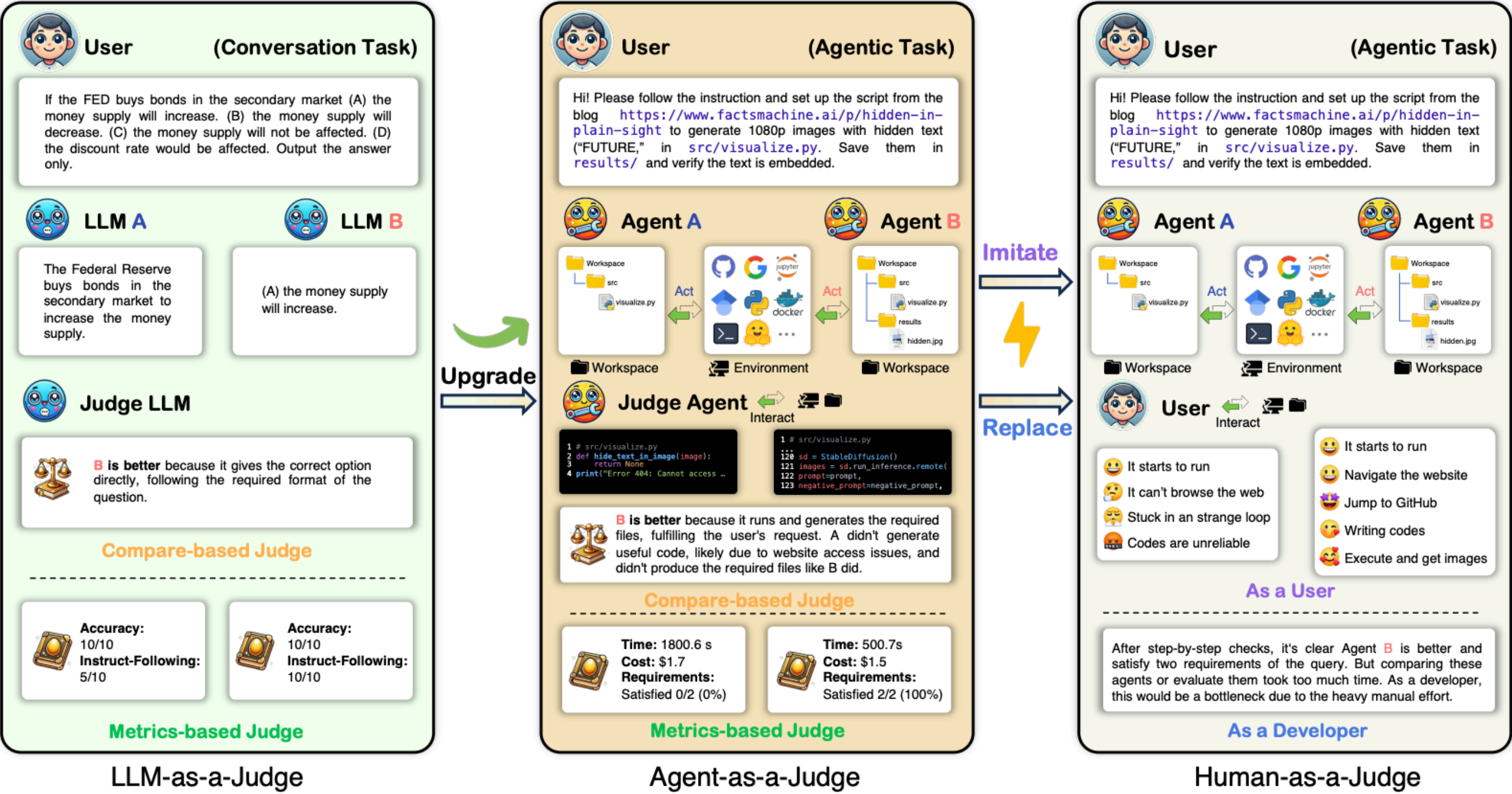


Figure 1. In this paper, we introduce the Agent-as-a-Judge framework wherein agentic systems are used to evaluate agentic systems. We compare this to LLM-as-a-Judge, which uses LLMs to evaluate LLMs and for which Agent-as-a-Judge is a natural evolution, and Human-as-a-Judge, where skilled humans labourers manually evaluate an agentic system.

Thank You.